InsightAI Analytics Report

Automated Exploratory Data Analysis & Quality Audit Summary

1. Executive Summary

InsightAI has processed the dataset containing **3 rows** and **5 columns**. This dataset consists of **2 numeric** variables and **3 categorical** variables. Overall, the data health is **Excellent** with **0.00% missing entries** and **0.00% duplicate records**. Several notable distributions and correlation dependencies have been analyzed below.

Dataset High-Level Metrics

Metric	Value
Total Rows (Records)	3
Total Columns (Variables)	5
Memory Occupancy	0.42 KB
Missing Data Cells	0 (0.0%)
Duplicate Rows	0 (0.0%)

2. Attributes Schema Details

This table details the variable names, data types, missing ratios, and uniqueness checks calculated across columns.

Column Name	Type	Storage DType	Missing %	Uniques
Name	Text	object	0.0%	3
Age	Numeric	float64	0.0%	3
Salary	Numeric	float64	0.0%	3
Join Date	DateTime	datetime64[ns]	0.0%	3
Country	Text	object	0.0%	3

3. Data Quality & Next Steps

Quality Issues Checklist

- **No severe quality issues detected!** The dataset structure is clean, types are correct, and there are no missing cells or duplicates.

Strategic Actions & Recommendations

5. **Reduce Multicollinearity**: Pairs like 'Age' & 'Salary' are highly correlated. Consider dropping one to prevent model overfitting.
6. **One-Hot Encoding**: Apply One-Hot encoding to nominal variables before model training to create proper numeric model inputs.

4. Linear Relationship & Statistical Correlations

The following linear associations were found to be statistically significant: - **'Age'** and **'Salary'**: Share a **moderate positive linear relationship** (Pearson r = **0.65**). *Insight: Strong positive correlations suggest these variables move together; they may be redundant in machine learning models.*

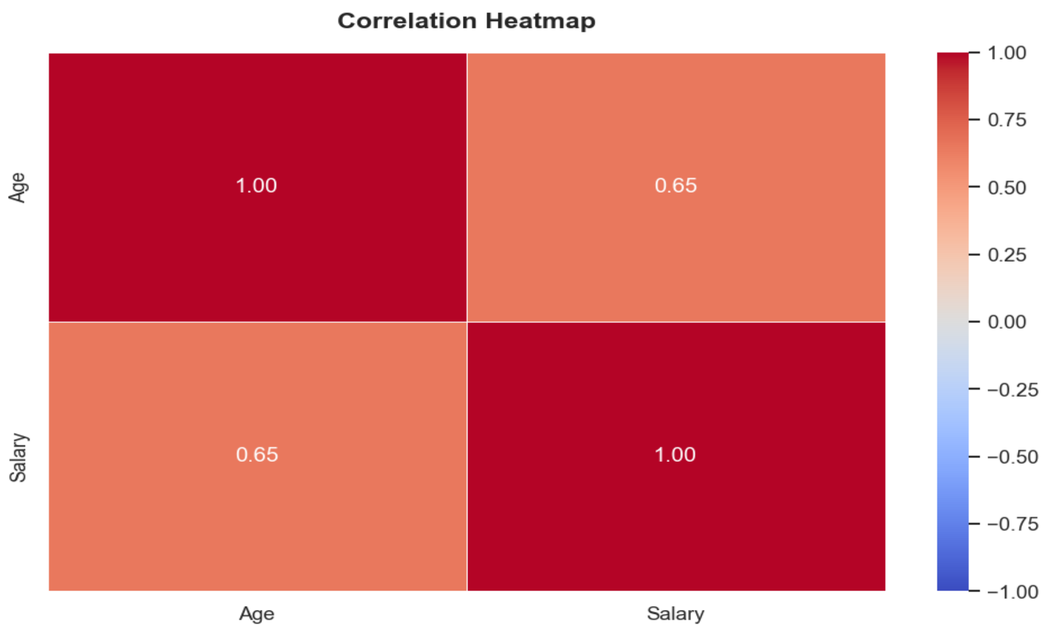

Figure 4.1: Correlation Coefficient Matrix (Pearson Correlation R)

5. Outliers & Missingness Analysis

No outliers were detected under standard threshold limits. All data points lie within normal variance bounds.

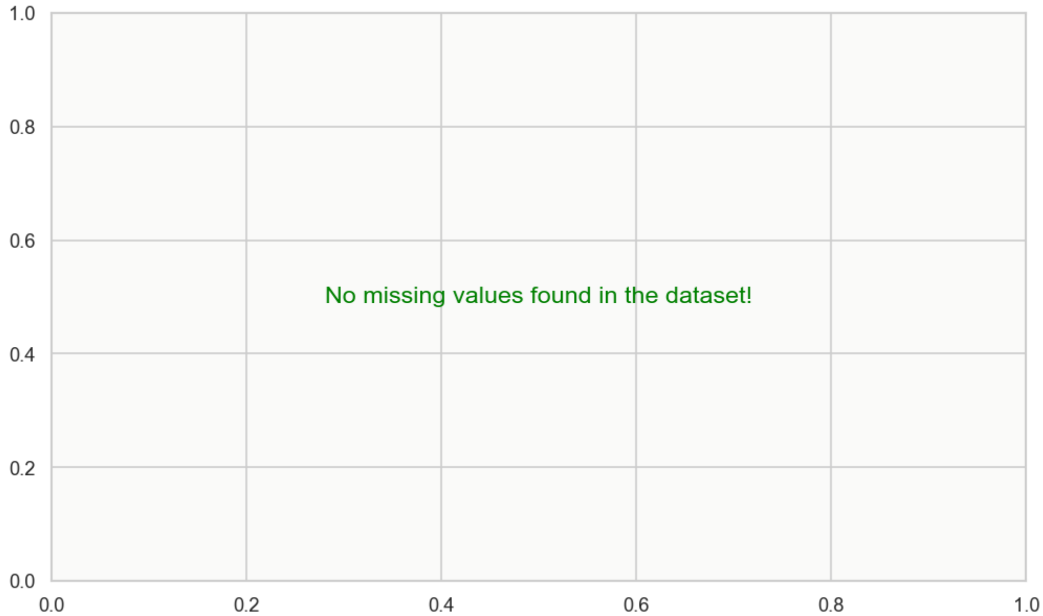

Figure 5.1: Missing Values Heatmap (Correlations of nulls)